

Writing Your Senior Project Report with LaTeX and Overleaf

A Complete Student Guide to the APA 7 LaTeX Template

Senior Project I & II

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Download the template:

github.com/ParisaRostaami/APA7-LaTeX-Template

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BEFORE YOU BEGIN

About This Guide

This guide teaches you everything you need to write your Senior Project report in APA 7 style using LaTeX and Overleaf. You do not need any previous experience with LaTeX. We start with what LaTeX is and why it is worth learning, move through the basics of Overleaf, and then walk through every part of the APA 7 student template that you will use for your report.

What You Will Be Able to Do

By the end of this guide, you will be able to:

- explain what LaTeX and Overleaf are and how a LaTeX document becomes a PDF;
- create, edit, compile, and download a LaTeX project in Overleaf;
- upload the APA 7 template and fill in a correct APA student title page;
- organize your report with APA heading levels;
- manage sources in a `.bib` file and cite them correctly in APA style;
- add APA-formatted tables, figures, equations, code excerpts, and appendices;
- structure a complete Senior Project report and check it before submission; and
- find and fix the most common LaTeX errors on your own.

How This Guide Is Organized

Part	Chapters	What it covers
I. Foundations	1–3	What LaTeX and Overleaf are, and the LaTeX basics you need
II. The APA 7 Template	4–11	APA rules, getting the template, and every element of the template in detail
III. Your Report	12–13	Structuring your Senior Project report, teamwork, the submission checklist, and troubleshooting
Appendices	A–D	Cheat sheet, working on your own computer with Git, exercise summary, and resources

Conventions Used in This Guide

You see	It means
<code>\textbf{text}</code>	Something you type exactly as shown, in Overleaf or in a file.
File → Word Count	A menu, button, or option to click in Overleaf.
Try It box (green)	A hands-on exercise. We will do these together in class.

You see	It means
Tip box (blue)	A shortcut or good habit that saves time.
Watch Out box (orange)	A common mistake and how to avoid it.
Key Idea box (navy)	A concept you should remember.

Key Idea: In a hurry? The five-step quick start

1. Download `APA7_Student_Template.zip` from the GitHub repository (Chapter 5).
2. In Overleaf, click **New Project** → **Upload project** and select the zip file. Do not unzip it first.
3. Open `main.tex` and fill in the title page block (Chapter 6).
4. Write your report, cite sources with `\parencite{key}`, and add entries to `references.bib` (Chapter 8).
5. Click **Recompile**, check the PDF against the checklist in Chapter 12, and download it.

PART I · FOUNDATIONS

1. What Is LaTeX?

1.1 The Idea in One Sentence

LaTeX (usually pronounced “LAY-tek” or “LAH-tek”) is a system for preparing documents: you write plain text mixed with short formatting instructions, and a program turns that text into a polished PDF.

In Microsoft Word or Google Docs, you format by clicking buttons and you see the final look as you type. This is called *WYSIWYG*, “what you see is what you get.” LaTeX works differently. You describe the *structure* of your document (“this is a section heading,” “this is a citation,” “this word is emphasized”), and LaTeX decides how it should look by following a set of rules. This approach is sometimes called *WYSIWYM*, “what you see is what you mean.”

What you type (main.tex)

```
\section*{Results}
Our model reached an accuracy of 92\% on the
\textbf{test set}, which is \emph{higher} than
the baseline. Three things helped:
\begin{itemize}
  \item data augmentation,
  \item early stopping, and
  \item a smaller learning rate.
\end{itemize}
The loss function was
\[ L = -\sum_{i=1}^n y_i \log \hat{y}_i \]
```

What you get (PDF)

Results

Our model reached an accuracy of 92% on the **test set**, which is *higher* than the baseline. Three things helped:

- data augmentation,
- early stopping, and
- a smaller learning rate.

The loss function was

$$L = -\sum_{i=1}^n y_i \log \hat{y}_i$$

Figure 1. On the left is what you type in LaTeX. On the right is the PDF that LaTeX produces from it.

Look closely at Figure 1. You never told LaTeX which font size to use for “Results,” how far to indent the bullet points, or how to draw the summation sign. You only said *what* each piece is, and LaTeX handled the *how*. That is the core idea. For your report, the APA 7 template contains all of the “how” rules, so you only need to write the “what.”

1.2 LaTeX Compared With a Word Processor

	Word or Google Docs	LaTeX
How you format	Click buttons; the result appears immediately	Type short commands; compile to see the result

	Word or Google Docs	LaTeX
Consistency	Manual; easy for headings and spacing to drift	Automatic; every heading follows the same rule
Citations and references	Typed by hand or with a plug-in	Generated automatically from your list of sources
Equations	Equation editor	Excellent, the standard in science and engineering
Long documents	Can become slow and unstable	Handles hundreds of pages without trouble
Teamwork and version control	Binary files that are hard to compare	Plain text that works well with Git
Learning curve	Low	Moderate for the first few hours, then easy

1.3 Why We Use LaTeX for the Senior Project

- **The template does the APA formatting for you.** Margins, spacing, fonts, headings, the title page, and the reference list are all handled automatically. You focus on content.
- **Citations cannot drift out of sync.** If you cite a source, it appears in the reference list. If you remove the citation, the entry disappears. No more missing or extra references.
- **It handles technical content well.** Equations, algorithms, code excerpts, and tables of results look professional with very little effort.
- **It is a career skill.** Most computer science research papers, theses, and many technical reports are written in LaTeX. If you plan to go to graduate school or work in research, you will use it again.
- **It is plain text.** Your report can live in a Git repository next to your code, and teammates can see exactly what changed.

1.4 How LaTeX Turns Your Text Into a PDF

Writing a LaTeX document is a lot like writing a program. You write *source code* (a file ending in `.tex`), and a *compiler* builds the output (a PDF). When you click **Recompile** in Overleaf, a short sequence of programs runs behind the scenes, as shown in Figure 2.

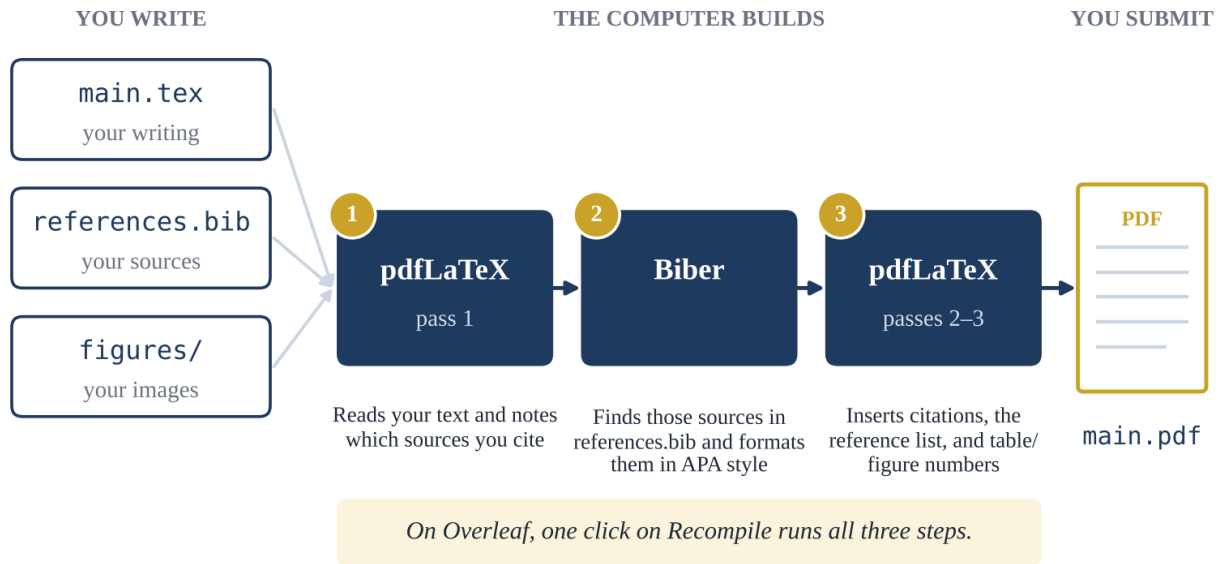


Figure 2. What happens when you click Recompile. Overleaf runs all of these steps for you.

Just like code, a LaTeX document can contain errors. If there is an error, LaTeX tells you which line it is on, and it may not produce a PDF until you fix it. Chapter 13 shows you how to read these messages. Do not worry: most errors are simple typos, such as a missing brace.

1.5 Key Vocabulary

You will see these terms throughout the guide. You do not need to memorize them now; come back to this table whenever you need it.

Term	Meaning
Source file (<code>.tex</code>)	The plain-text file you write. Your report lives in <code>main.tex</code> .
Compile	Turning the source file into a PDF. In Overleaf you click Recompile .
Command	An instruction that starts with a backslash, such as <code>\section{Method}</code> .
Argument	The input a command needs, written inside braces: the <code>Method</code> in <code>\section{Method}</code> .
Option	An optional setting written in square brackets, such as <code>[width=0.5\linewidth]</code> .
Environment	A block that starts with <code>\begin{name}</code> and ends with <code>\end{name}</code> , such as a list or table.
Package	An add-on that gives LaTeX new features, loaded with <code>\usepackage{name}</code> .
Document class	The overall layout rules of a document. Our template uses the <code>apa7</code> class.
Preamble	Everything before <code>\begin{document}</code> : settings, packages, and title-page information.
Body	Everything between <code>\begin{document}</code> and <code>\end{document}</code> : the content of your paper.

Term	Meaning
Comment	Text after a % sign. LaTeX ignores it, so it never appears in the PDF.
Bibliography file (.bib)	The file that stores your sources. Ours is references.bib.
Citation key	The short, unique name you give a source, such as lecun2015deep.
Biber	The program that reads your .bib file and formats references in APA style.
Float	A table or figure that LaTeX is allowed to move to the best position on the page.
Log	The report LaTeX writes while compiling. It contains errors and warnings.

CHAPTER 2

2. What Is Overleaf?

2.1 Overview

Overleaf is a website for writing LaTeX documents. It gives you an editor, a compiler, and a PDF viewer in your web browser, so there is nothing to install. Your projects are saved automatically in the cloud, you can open them from any computer, and you can share them with teammates and your instructor.

For this course, Overleaf is the recommended way to work. If you prefer to work on your own computer, Appendix B explains how, but you do not need to.

2.2 Creating Your Account

1. Go to overleaf.com and click **Register** (or **Sign up**).
2. Sign up with your university email address. Using your school address keeps your coursework in one place and makes it easy for your instructor to recognize you when you share a project.
3. Open the confirmation email from Overleaf and click the link to verify your address.
4. Log in. You will land on your **Projects** dashboard, which lists all of your projects.

2.3 What the Free Plan Includes

The free Overleaf plan is enough for this course. A few limits are worth knowing, especially for team projects:

Feature	Free plan
Number of projects	Unlimited
Collaborators	One invited collaborator per project (see Section 12.6 for team strategies)
Compile timeout	10 seconds per compile. The template compiles in a few seconds.
Track changes and full project history	Paid plans only. The free plan keeps only recent history.
Git and GitHub integration	Paid plans only (you can still use Git on your own computer; see Appendix B)

These limits come from Overleaf's documentation at the time of writing (October 2026) and may change. Check the current plans at overleaf.com/user/subscription/plans.

2.4 A Tour of the Overleaf Editor

When you open a project, you see the editor. Figure 3 is a simplified drawing of it. The numbers match the table below.

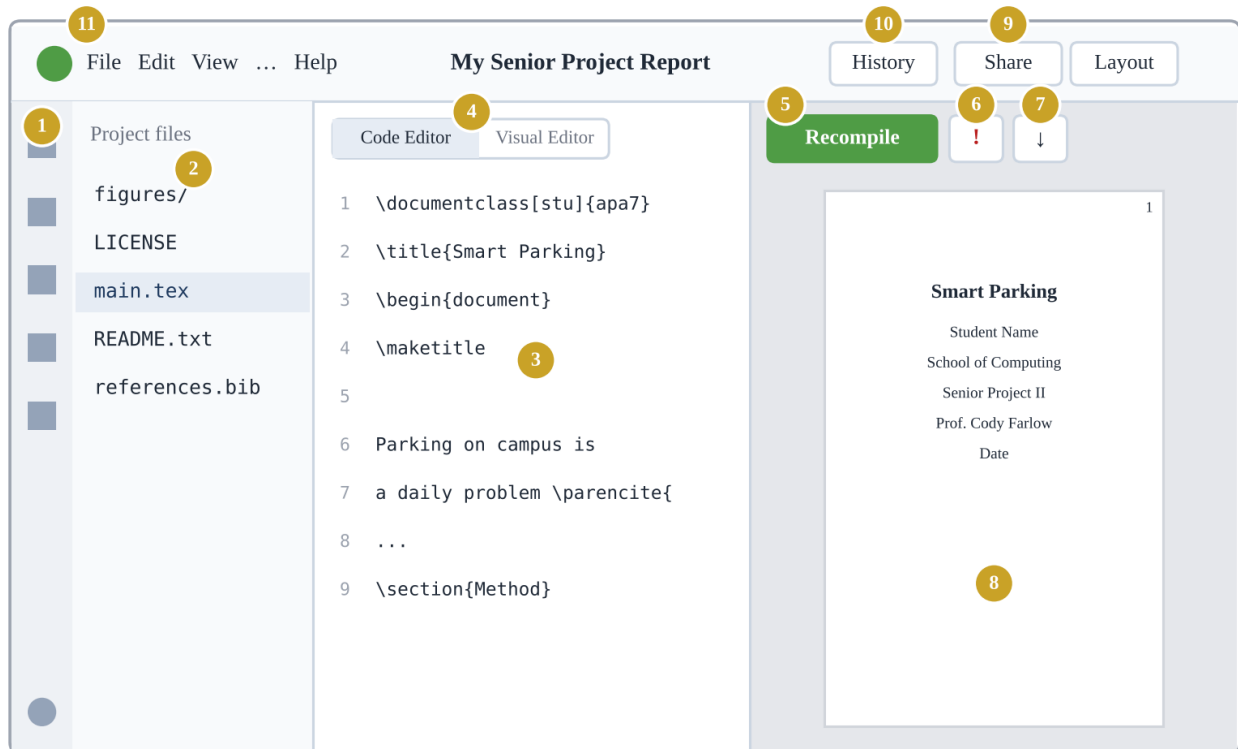


Figure 3. The main parts of the Overleaf editor (simplified drawing; your screen may look slightly different as Overleaf updates its design).

#	Part	What it does
1	Left rail	Icons for the file tree, search, integrations, review, and chat. Settings and Help are at the bottom.
2	File tree	All files in your project. Click a file to open it. Buttons at the top create files and folders and upload files.
3	Editor	Where you type. Line numbers help you find errors.
4	Code Editor / Visual Editor	Switches between raw LaTeX (Code Editor) and a word-processor-like view (Visual Editor). This guide uses the Code Editor.
5	Recompile	Builds the PDF from your source. The small arrow next to it has options such as auto-compile.
6	Logs and output files	Shows errors (red), warnings (yellow), and the raw log. A number appears here when something is wrong.
7	Download PDF	Saves the compiled PDF to your computer.
8	PDF preview	The compiled document. Double-click any spot to jump to the matching line in the editor.

#	Part	What it does
9	Share	Invite collaborators or create a link to your project.
10	History	See and restore earlier versions (limited on the free plan).
11	Menu bar	File , Edit , View , and Help menus. For example, File → Word Count and File → Download .

2.5 Features That Save Time

Action	Windows	Mac
Recompile	Ctrl + Enter or Ctrl + S	Cmd + Enter or Cmd + S
Comment or uncomment selected lines	Ctrl + /	Cmd + /
Find and replace	Ctrl + F	Cmd + F
Make selected text bold or italic	Ctrl + B / Ctrl + I	Cmd + B / Cmd + I
Undo	Ctrl + Z	Cmd + Z

- **Jump between PDF and code.** Double-click text in the PDF preview to jump to its source line. This is the fastest way to find where something is written.
- **Auto-compile.** In the menu next to **Recompile** you can turn on auto-compile so the PDF updates as you type. Many people prefer to recompile manually.
- **Spell check.** Overleaf underlines misspelled words. You can set the language in **Settings**.

Try It: 1 — Your first LaTeX document (10 minutes)

1. In your Overleaf dashboard, click **New Project** → **Blank project** and name it **LaTeX Practice**.
2. Overleaf opens a short example document. Select everything in `main.tex` (Ctrl + A or Cmd + A) and delete it.
3. Type the code below exactly as shown (Figure 4 shows the code and the result).
4. Click **Recompile**.

```
\documentclass{article}
\title{My First \LaTeX{} Document}
\author{Your Name}
\date{\today}

\begin{document}
\maketitle

\section*{Introduction}
Hello, world! This is my first document
written in \LaTeX. I can make text
\textbf{bold} and \textit{italic}.
\end{document}
```

Checkpoint: The PDF preview shows a centered title, your name, today’s date, an “Introduction” heading, and one paragraph with a bold word and an italic word.

main.tex

```
\documentclass{article}
\title{My First \LaTeX{} Document}
\author{Student Name}
\date{\today}

\begin{document}
\maketitle

\section*{Introduction}
Hello, world! This is my first document
written in \LaTeX. I can make text
\textbf{bold} and \textit{italic}.
\end{document}
```

PDF output

My First L^AT_EX Document

Student Name
October 2, 2026

Introduction

Hello, world! This is my first document written in L^AT_EX. I can make text **bold** and *italic*.

Figure 4. The code for Try It 1 and the PDF it produces.

CHAPTER 3

3. LaTeX Basics You Need

This chapter covers the small set of LaTeX ideas you need for your report. You do not need to learn all of LaTeX: the template takes care of most of it.

3.1 The Anatomy of a LaTeX Document

Every LaTeX document has the same three parts:

```
\documentclass{article}      % 1. The document class: what kind of document

\usepackage{graphicx}        % 2. The preamble: packages and settings

\begin{document}             % 3. The body: everything that appears in the PDF
Your text goes here.
\end{document}
```

- The **document class** comes first and sets the overall design. Our template uses `\documentclass[...]{apa7}`.
- The **preamble** is everything between the document class and `\begin{document}`. It loads packages and stores settings.
- The **body** is everything between `\begin{document}` and `\end{document}`. Anything after `\end{document}` is ignored.

3.2 Commands

A command starts with a backslash (`\`) followed by letters. Many commands take an **argument** in braces `{ }`, and some accept **options** in square brackets `[ ]`.

Example	What it does
<code>\textbf{important}</code>	Prints “important” in bold. The argument is the text to format.
<code>\section{Method}</code>	Starts a section with the heading “Method.”
<code>\includegraphics[width=0.5\linewidth]{figures/ui.png}</code>	Inserts an image. The option sets its width; the argument is the file name.
<code>\maketitle</code>	A command with no argument. It prints the title page.

Watch Out: Commands are case-sensitive

`\Section{Method}` and `\textBF{word}` will cause errors. Always type commands in lowercase unless this guide shows otherwise.

3.3 Environments

An environment applies a format to a block of content. It always has a matching pair of `\begin{...}` and `\end{...}` with the same name:

```
\begin{itemize}
  \item First point
  \item Second point
\end{itemize}
```

If you forget the `\end{...}` line, or misspell the name in one of the two lines, you will get an error. Indenting the lines inside an environment is optional, but it makes your code easier to read.

3.4 Paragraphs and Spacing

- **A blank line starts a new paragraph.** This is the only way you should create paragraphs.
- **A single line break does nothing.** LaTeX joins the lines into the same paragraph. You can put each sentence on its own line, which makes editing and teamwork easier.
- **Several spaces count as one space.** You cannot add extra space by pressing the space bar many times.
- **Do not fix spacing by hand.** Avoid `\\` to end paragraphs and avoid `\vspace` to add gaps. The APA template already sets double spacing and indentation correctly.

This is the first paragraph. It continues
on this line because there is no blank line.

This is a second paragraph, because a blank line came before it.

3.5 Special Characters

Ten characters have a special meaning in LaTeX. To print them as ordinary characters, type the version in the second column. The first five are the ones students run into most often.

Character	Type this	Why it matters
%	<code>\%</code>	<code>%</code> starts a comment. Without the backslash, the rest of the line disappears silently! Write <code>85\%</code> .
&	<code>\&</code>	<code>&</code> separates table columns. Write <code>R\&D</code> or Johnson <code>\&</code> Johnson.
_	<code>_</code>	<code>_</code> is for subscripts in math. Write <code>my_variable</code> or <code>file_name.py</code> .

Character	Type this	Why it matters
#	<code>\#</code>	Used internally by commands. Write <code>C\#</code> or issue <code>\#12</code> .
\$	<code>\\$</code>	<code>\$</code> starts and ends math. Write <code>\\$500</code> .
{ }	<code>\{ \}</code>	Braces group arguments.
~	<code>\textasciitilde{}</code>	<code>~</code> is a non-breaking space (see Section 3.6).
^	<code>\textasciicircum{}</code>	<code>^</code> is for superscripts in math.
\	<code>\textbackslash{}</code>	Starts every command. Note: <code>\\</code> means “new line,” not a backslash.

Tip: URLs and file paths

Inside `\url{...}`, special characters such as `_`, `%`, and `#` are handled for you: `\url{https://github.com/team/my_repo}` works as is.

3.6 Formatting Text

You type	You get	Notes
<code>\textbf{bold}</code>	bold	For strong emphasis; use sparingly in APA papers.
<code>\textit{italic}</code>	<i>italic</i>	For titles of books, key terms on first use, and statistics.
<code>\emph{emphasis}</code>	<i>emphasis</i>	Preferred for emphasis; LaTeX chooses the right style.
<code>\texttt{print()}</code>	<code>print()</code>	Monospaced text for code, file names, or commands.
<code>`quoted'</code>	“quoted”	Two backticks, then two apostrophes. Or use <code>\enquote{quoted}</code> .
pages 10--20	pages 10–20	Two hyphens make an en dash, used for ranges.
a pause---like this	a pause—like this	Three hyphens make an em dash.
Table~1, p.~5	Table 1, p. 5	<code>~</code> is a space that never breaks across lines.
and so on\ldots	and so on...	Correctly spaced ellipsis.

Watch Out: Straight quotation marks

Typing the " key produces wrong-looking quotation marks in LaTeX. Use `` and '' (or \enquote{...}) so the PDF shows proper curly quotation marks.

3.7 Lists

Use `itemize` for bullet points and `enumerate` for numbered lists. Lists can be nested.

```
\begin{enumerate}
  \item Collect the parking sensor data.
  \item Clean the data:
    \begin{itemize}
      \item remove duplicate readings, and
      \item fill in missing timestamps.
    \end{itemize}
  \item Train the prediction model.
\end{enumerate}
```

APA allows bulleted lists and numbered lists. Use a numbered list when the order matters (for example, the steps of a procedure).

3.8 Math

Math goes between dollar signs for *inline* math, which sits in the sentence, or inside an `equation` environment for a *displayed*, numbered equation.

The model has n layers and a learning rate of $\alpha = 0.001$.

```
\begin{equation}
  \text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}
  \label{eq:accuracy}
\end{equation}
```

You type (inside math)	Meaning	You type (inside math)	Meaning
x^2	superscript	$\leq \geq \neq$	$\leq \geq \neq$
x_i	subscript	$\times \cdot$	$\times \cdot$
$\frac{a}{b}$	fraction	$\in \rightarrow$	$\in \rightarrow$
$\sqrt{x}$	square root	$\alpha \beta \theta$	$\alpha \beta \theta$
$\sum_{i=1}^n$	summation	$\log \max$	$\log \max$
word	normal text in math	$O(n \log n)$	$O(n \log n)$

Tip: Finding a symbol

If you do not know the command for a symbol, draw it at detexify.kirelabs.org, and it will suggest the command.

3.9 Comments

Anything after `%` on a line is a comment. Comments are useful for notes to yourself or your team, and for temporarily hiding text without deleting it. Select lines and press `Ctrl + /` (`Cmd + /` on a Mac) to comment or uncomment them.

```
% TODO (Jordan): add the latency results here
We tested the system with 50 users. % this sentence will appear
% This whole sentence is hidden from the PDF.
```

3.10 Reading Errors: The First Look

When something is wrong, a red number appears next to the **Logs and output files** icon. Click it to see the messages. Each error says what went wrong and on which line. Three habits will solve most problems:

1. **Fix the first error first.** One mistake can cause a chain of later errors that disappear once the first is fixed.
2. **Look at the reported line and the line just before it.** Missing braces are often on the previous line.
3. **Recompile after each fix.** Small steps make it easy to see what helped.

Chapter 13 lists the most common error messages and how to fix each one.

Try It: 2 — Practice the basics (15 minutes)

In your `LaTeX Practice` project, add the following below your Introduction paragraph:

- a new section with `\section*{Practice}`;
- a sentence that uses the characters `%`, `&`, and `_` correctly (for example, “Our R&D budget grew by 15% in file_v2.”);
- a numbered list with three items, one of which contains a nested bullet list; and
- one inline equation and one numbered equation.

Now break something on purpose: delete one closing brace `}` and click **Recompile**. Open the logs, read the error message, find the line it points to, and fix it.

Checkpoint: Your PDF shows the special characters correctly, both lists, and an equation numbered (1) on the right. You found and fixed the error you created.

PART II · THE APA 7 TEMPLATE

4. APA 7 Student Paper Essentials

4.1 What APA Style Is

APA style is a set of rules for formatting and writing papers, published by the American Psychological Association in the *Publication Manual of the American Psychological Association* (7th ed., 2020). It covers how a paper looks (margins, fonts, headings), how you credit sources (in-text citations and the reference list), and how you write (clear, concise, bias-free language).

APA 7 has two versions of the paper format: the *professional paper* (for journal submissions) and the *student paper* (for coursework). **Your report is a student paper.** The template is set up for the student format.

Two trustworthy references for APA rules are the [Purdue OWL APA General Format page](#) and the official [APA Style website](#).

4.2 The Rules, and Who Takes Care of Each One

The table below lists the main APA 7 student paper rules. The last column tells you whether the template handles the rule for you or whether it is your job.

Requirement	APA 7 rule (student paper)	Who handles it
Page and margins	US Letter (8.5 × 11 in.), 1-in. margins on all sides	Template
Font	A readable font used consistently, such as Times New Roman 12 pt (Calibri 11, Arial 11, Lucida Sans Unicode 10, Georgia 11, and Computer Modern 10 are also accepted)	Template (Times New Roman style)
Line spacing	Double-spaced throughout, including the title page and references	Template
Alignment	Flush left with a ragged right edge (not justified)	Template
Paragraph indent	First line of each paragraph indented 0.5 in.	Template
Page numbers	Top right corner of every page, starting with the title page	Template
Running head	Not required for student papers	Template (omitted)
Title page	Title, author name(s), affiliation, course number and name, instructor, due date	You fill in the fields; the template lays them out
Abstract	Only if your instructor asks for one	You decide (keep or remove)
Headings	Five levels with specific formats	You choose the level; the template formats it

Requirement	APA 7 rule (student paper)	Who handles it
Tables and figures	Bold number, italic title, notes below	You write the title; the template formats it
In-text citations	Author–date format, such as (LeCun et al., 2015)	You pick the command; the template formats it
Reference list	New page, “References” centered in bold, alphabetical, hanging indent	Automatic, from <code>references.bib</code>
Capitalization	Title case for the paper title and headings; sentence case for titles in the reference list	You (title case); Biber helps with references
Writing style	Clear, precise, bias-free; consistent tense; numbers and abbreviations used correctly	You (see Section 12.5)

Key Idea: The template formats; you write

The template guarantees the *look* of an APA paper. Following APA is also about *content*: an accurate and complete `.bib` entry for every source, headings that match the structure of your report, and honest attribution of everything you did not create yourself.

4.3 Title Case and Sentence Case

Two capitalization styles appear throughout APA papers:

Style	Rule	Example	Used for
Title case	Capitalize major words and any word of four or more letters. Lowercase short articles, conjunctions, and prepositions (a, an, the, and, but, for, of, to, in, on, at, by) unless they start the title or subtitle.	A Real-Time Parking Availability System for Urban Campuses	Paper title, headings, table and figure titles
Sentence case	Capitalize only the first word, the first word after a colon, and proper nouns.	A real-time parking availability system for urban campuses	Titles of works in the reference list (Biber applies this for you)

CHAPTER 5

5. Getting the Template into Overleaf

5.1 Download the Template From GitHub

The template is published in a public GitHub repository. You do not need a GitHub account to download it.

What	Where
Repository page	github.com/ParisaRostaami/APA7-LaTeX-Template
Direct download of the template	APA7_Student_Template.zip
QR code	On the cover of this guide

To download it from the repository page:

1. Open the repository page.
2. In the list of files, click `APA7_Student_Template.zip`.
3. Click the **Download raw file** button (the downward-arrow icon on the right side of the file view).
4. The file `APA7_Student_Template.zip` is saved to your **Downloads** folder.

Watch Out: Use the template zip, not “Download ZIP”

The green **Code** → **Download ZIP** button on the repository page downloads *everything* in the repository, including this guide. For Overleaf, download only `APA7_Student_Template.zip` as described above.

Watch Out: Mac users: Safari may unzip the file automatically

If you find a folder named `APA7_Student_Template` instead of a `.zip` file, Safari unzipped it for you. Either right-click the folder and choose **Compress**, or in Safari open **Settings** → **General** and uncheck **Open “safe” files after downloading**, then download again.

5.2 Upload the Template to Overleaf

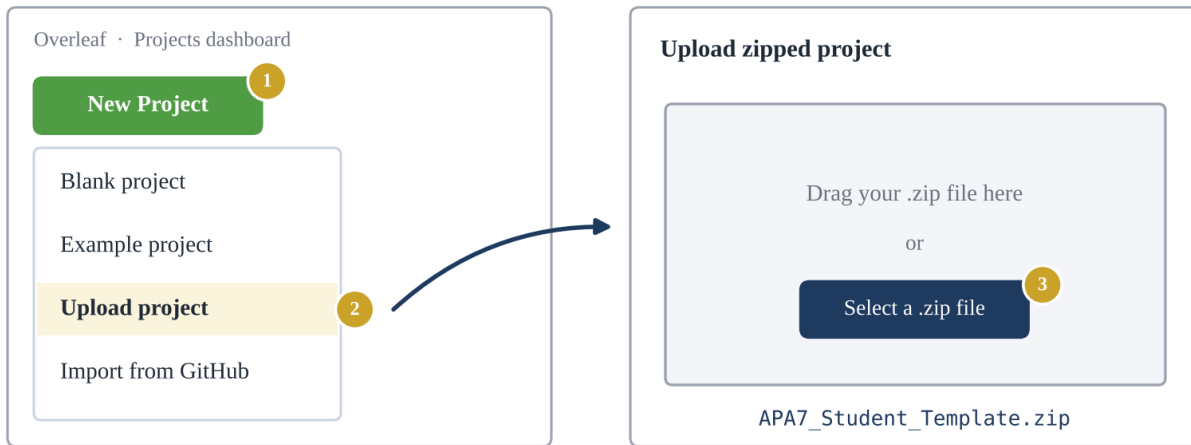


Figure 5. Creating a new Overleaf project from the template zip file. The numbers match the steps below.

1. Go to your Overleaf **Projects** dashboard and click **New Project**.
2. Choose **Upload project**.
3. Click **Select a .zip file** (or drag the file into the box) and choose `APA7_Student_Template.zip`. **Do not unzip it first.**
4. Overleaf creates the project and opens it in the editor. The project name comes from the zip file name.
5. Rename the project to something meaningful, for example `Team Phoenix – Senior Project I Report`. Click the project name at the top of the editor (or use **Rename** on the dashboard).

5.3 The First Compile

Overleaf usually compiles the project as soon as it opens. If the PDF preview is empty, click **Recompile**. After a few seconds, you should see a six-page sample paper: a title page, an abstract page, the main text with sample headings, a table and a figure, the reference list, and an appendix.

If something looks wrong, check two settings. Click **Settings** at the bottom of the left rail and open the **Compiler** tab:

Setting	Should be
Compiler	pdfLaTeX
Main document	<code>main.tex</code>
TeX Live version	Leave the default

Tip: You never run Biber yourself on Overleaf

Overleaf detects that the template uses Biber and runs it automatically every time you recompile (Figure 2). If references ever look stale, see “Clear cached files” in Section 13.1.

5.4 What Is Inside the Project

APA7_Student_Template/

├─ main.tex	— Your paper. This is where you write.
├─ references.bib	— Your list of sources (one entry per source).
├─ figures/	— Folder for your images.
│ └─ sample-figure.png	— Example image; replace or delete it.
├─ README.txt	— Quick-start notes. You can delete it.
└─ LICENSE	— Template license. Leave it as is.

Figure 6. The files in the template project and what each one is for.

You will spend almost all of your time in `main.tex` and `references.bib`. Images go in the `figures` folder. The `README.txt` file is a short version of this guide, and `LICENSE` explains the terms of use of the template. Neither affects your PDF.

Try It: 3 — Set up your report project (10 minutes)

1. Download `APA7_Student_Template.zip` from the repository.
2. In Overleaf, create a new project with **New Project** → **Upload project**.
3. Compile it and rename the project to `[Your Team Name] - Senior Project Report`.
4. In the file tree, click each file once to see what is inside.
5. In the PDF preview, double-click the heading “Tables and Figures” and notice that the editor jumps to the matching line in `main.tex`.

Checkpoint: You have your own copy of the template, it compiles into a six-page PDF, and you can move between the PDF and the source.

CHAPTER 6

6. A Guided Tour of main.tex

6.1 The Big Picture

Open `main.tex`. It follows the same structure as every LaTeX document (Section 3.1), with extra parts for the title page and references. Figure 7 shows the parts in order. Only the two highlighted parts need your attention: the title-page information and the body.

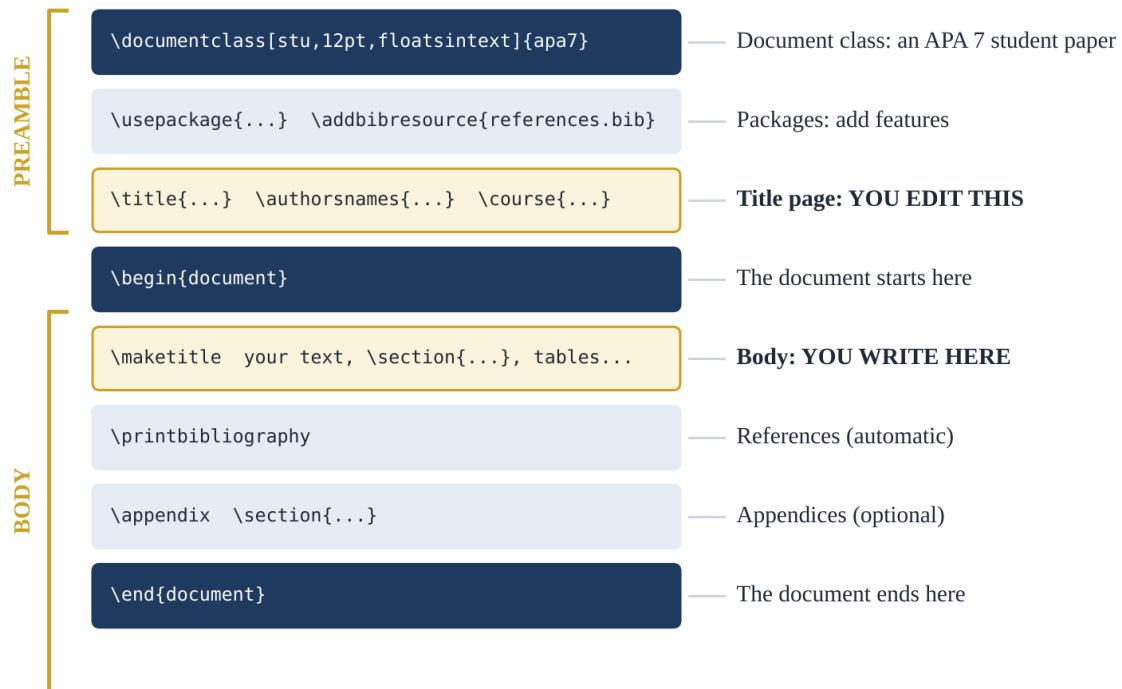


Figure 7. The structure of `main.tex`. You edit the highlighted parts.

The comments at the top of the file, starting with `%`, describe the template and its license. Leave them in place. They do not appear in your PDF.

6.2 The Document Class Line

```
\documentclass[stu,12pt,floatsintext]{apa7}
```

Option	What it does
<code>stu</code>	Student paper mode: creates the APA student title page (with course, instructor, and due date) and no running head.
<code>12pt</code>	Sets the font size to 12 points, the APA size for Times New Roman.
<code>floatsintext</code>	Places tables and figures in the text near where you mention them, instead of collecting them at the end of the paper.

Watch Out: Do not change “stu” to another mode

The `apa7` class also has `man`, `jou`, and `doc` modes for journal manuscripts. They add a running head and other features that student papers should not have.

6.3 The Packages

The next block loads the packages the template needs. You do not need to change anything here.

Line	Purpose
<code>\usepackage[american]{babel}</code>	American English rules for hyphenation and for words such as “and” in citations.
<code>\usepackage[T1]{fontenc}</code>	Correct handling of accented letters and special characters.
<code>\usepackage{newtxtext,newtxmath}</code>	Times New Roman–style text and matching math font.
<code>\usepackage{csquotes}</code>	Correct quotation marks; required by the APA citation style.
<code>\usepackage[style=apa,...]{biblatex}</code>	APA 7 citations and reference list, processed by Biber.
<code>\addbibresource{references.bib}</code>	Tells LaTeX where your sources are stored.
<code>\usepackage{graphicx}</code>	Lets you insert images with <code>\includegraphics</code> .
<code>\usepackage{booktabs}</code>	Clean horizontal table lines (<code>\toprule</code> , <code>\midrule</code> , <code>\bottomrule</code>).
<code>\usepackage{amsmath}</code>	Better equations.
<code>\hypersetup{hidelinks}</code>	Makes links and cross-references clickable without colored boxes.

Tip: Adding a package later

If you need a new feature, such as code listings (Section 11.2), add its `\usepackage` line in this block, before the title-page section.

6.4 Filling In the Title Page

Find the block marked **TITLE PAGE**. Replace each placeholder with your own information. Here is an example for a team of three:

```

\title{SmartPark: A Real-Time Parking Availability System for Campus Drivers}

\authorsnames{Alex Kim, Jordan Lee, Sam Rivera}

\authorsaffiliations{{School of Computing, Wichita State University}}

\course{CS 000: Senior Project I}
\professor{Prof. Cody Farlow}
\duedate{December 4, 2026}

```

Field	What to write	Rules and tips
<code>\title</code>	The full title of your report	Title case (Section 4.3). Be specific and descriptive. No period at the end. Avoid abbreviations the reader may not know. It is printed in bold automatically.
<code>\authorsnames</code>	Every team member's name	Separate names with commas. LaTeX adds “and” before the last name for you. Write names as first name, middle initial (optional), last name. Do not add titles or degrees.
<code>\authorsaffiliations</code>	Your department and university	Keep the double braces <code>{{...}}</code> . All team members share one affiliation.
<code>\course</code>	Course number and name	Use the course number and name exactly as they appear in the syllabus, for example <code>CS 000: Senior Project I</code> (replace <code>CS 000</code> with the real course number).
<code>\professor</code>	Your instructor	<code>Prof. Cody Farlow</code>
<code>\duedate</code>	The assignment due date	Write the month in full: <code>December 4, 2026</code> .

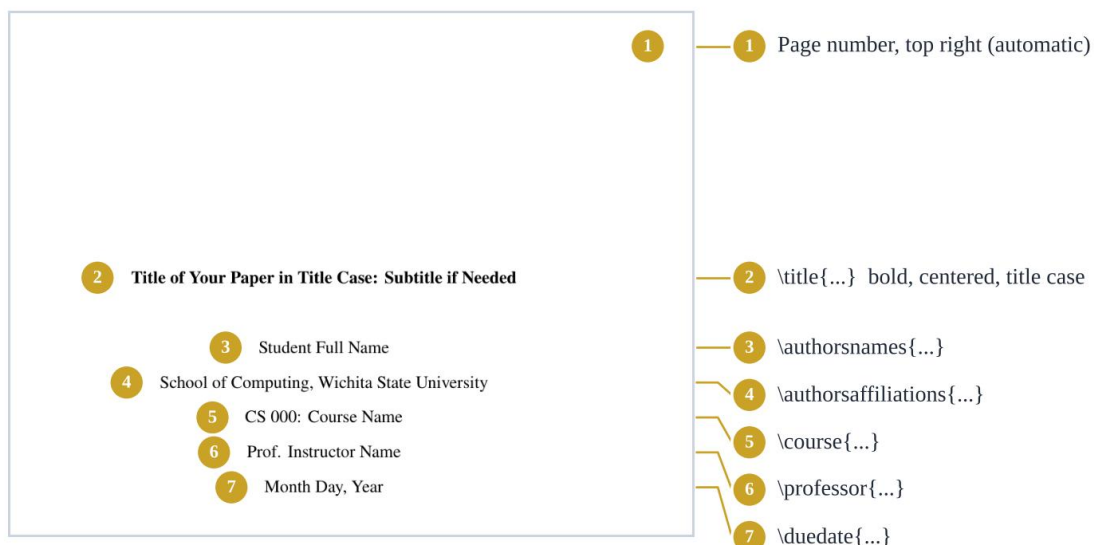


Figure 8. The title page of the template and the command that produces each line.

Watch Out: Keep the double braces

`\authorsaffiliations{{School of Computing, Wichita State University}}` must have *two* opening and *two* closing braces. With only one pair, the comma is treated as a separator and the affiliation is split incorrectly.

6.5 The Abstract and Keywords

APA does not require an abstract for student papers unless your instructor asks for one. **Follow your assignment instructions.**

- **To keep the abstract:** replace the placeholder text inside `\abstract{...}` with one paragraph, usually 150–250 words, that summarizes the problem, your approach, the main results, and why they matter. Replace the keywords with three to five terms that describe your project.
- **To remove the abstract:** put `%` at the start of the `\abstract` and `\keywords` lines (select them and press Ctrl + / or Cmd + /). The abstract page disappears from the PDF.

```
% \abstract{...}  
% \keywords{...}
```

6.6 Writing the Body

Your paper starts right after `\maketitle`. The template repeats your title in bold at the top of the first page of text, as APA requires. **Start writing your introduction directly below «\maketitle», without an “Introduction” heading;** in APA style, the first paragraphs after the title are understood to be the introduction.

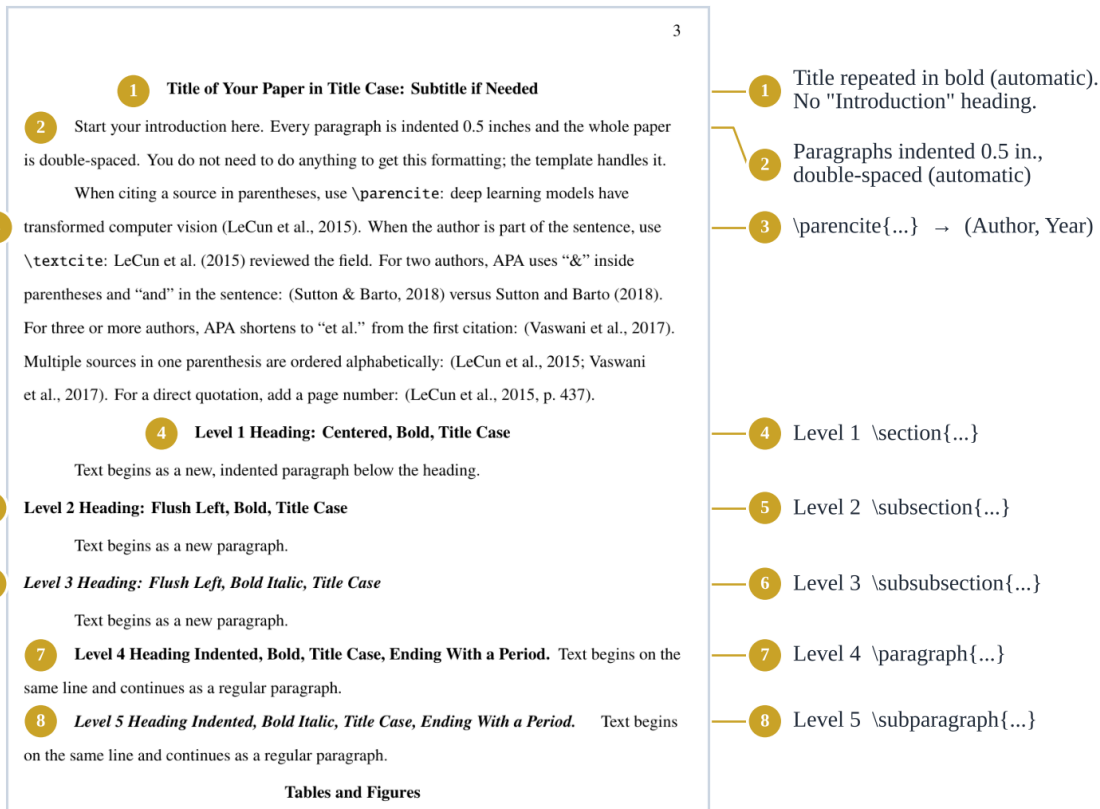


Figure 9. The first page of text. Everything marked “automatic” is handled by the template.

Delete the sample text and headings as you replace them with your own, but keep these lines:

```
\begin{document}
\maketitle
... your report ...
\printbibliography
... appendices (optional) ...
\end{document}
```

6.7 Optional: One File per Section

For a long report or a team, it is helpful to put each major section in its own file. Create a folder named `sections` in Overleaf (**New folder** button in the file tree), add a file for each section, and include them in `main.tex` with `\input`:

```
\maketitle
\input{sections/introduction}
\input{sections/requirements}
\input{sections/design}
\input{sections/testing}
\printbibliography
```

The section files contain only text and LaTeX commands, with no `\documentclass` and no `\begin{document}`. Each teammate can own one file, which reduces the chance of two people editing the same lines.

Try It: 4 — Make the title page yours (10 minutes)

- Fill in `\title`, `\authorsnames`, `\authorsaffiliations`, `\course`, `\professor`, and `\duedate` for your team.
- Decide with your team whether your assignment needs an abstract. Keep it, or comment it out.
- Recompile and compare your title page with Figure 8.

Checkpoint: Your title page shows your real title in bold title case, all team members joined with “and,” your affiliation, the course, Prof. Cody Farlow, and the due date. The page number 1 appears at the top right.

CHAPTER 7

7. Headings and Structure

7.1 The Five APA Heading Levels

APA 7 has five levels of headings. Each level has its own format, and each one maps to a LaTeX command. You only choose the level; the template applies the format.

Level	APA format	LaTeX command
1	Centered, Bold, Title Case	<code>\section{...}</code>
2	Flush Left, Bold, Title Case	<code>\subsection{...}</code>
3	Flush Left, Bold Italic, Title Case	<code>\subsubsection{...}</code>
4	Indented, Bold, Title Case, Ending With a Period. Text continues on the same line.	<code>\paragraph{...}</code>
5	Indented, Bold Italic, Title Case, Ending With a Period. Text continues on the same line.	<code>\subparagraph{...}</code>

In most reports, you will use Levels 1 to 3. Here is how they look in source code:

```
\section{System Design}
The system has three components ...

\subsection{Architecture}
We chose a client-server architecture because ...

\subsubsection{Database Layer}
The database stores ...

\paragraph{Indexing Strategy}
Write the text right after the heading command; it continues on the same line.
```

7.2 Heading Rules

- **No “Introduction” heading.** The introduction begins right after the title (Section 6.6).
- **No numbers.** APA headings are not numbered, and the template does not number them. Do not type numbers into the heading text.
- **Do not skip levels.** A Level 3 heading must sit under a Level 2 heading.
- **Use at least two subsections, or none.** If a section has a Level 2 heading, it should have at least two of them.
- **Use title case** for every heading (Section 4.3).
- **Level 4 and 5 headings end with a period**, which the template adds automatically. Do not type one yourself.

Tip: Page breaks

The reference list and each appendix start on a new page automatically. You rarely need to force a page break; if you must, use `\clearpage`.

CHAPTER 8

8. Citations and References

This is the most important chapter. Correct citations are part of academic integrity, and they are where LaTeX saves you the most time.

8.1 How It Works

Citations in the template follow a simple three-part system:

1. **Store each source once** as an entry in `references.bib`. Each entry has a unique **citation key**.
2. **Cite the source** in `main.tex` by its key, for example `\parencite{lecun2015deep}`.
3. **Recompile**. Biber looks up the key, formats the in-text citation, and adds the source to the reference list in APA style.

1 In main.tex you write

```
Deep learning changed computer
vision
\parencite{lecun2015deep}.
```

2 In references.bib you saved

```
@article{lecun2015deep,
  author = {LeCun, Yann and
    Bengio, Yoshua and
    Hinton, Geoffrey},
  title = {Deep learning},
  journal = {Nature}, year = {2015},
```

same key

Biber formats it in APA

3 In the PDF you get

In the text: Deep learning changed computer vision (LeCun et al., 2015).

References: LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep learning. *Nature*, 521(7553), 436–444.
<https://doi.org/10.1038/nature14539>

Figure 10. From citation key to formatted citation and reference list entry.

Key Idea: Only cited sources appear in the reference list

An entry in `references.bib` that you never cite will *not* appear in your references. This is exactly what APA requires: every cited source is in the reference list, and every source in the list is cited. You can keep extra entries in the file without worrying.

8.2 Anatomy of a .bib Entry

```
@article{lecun2015deep,          <- entry type and citation key
  author = {LeCun, Yann and Bengio, Yoshua and
            Hinton, Geoffrey},
  title  = {Deep learning},
  journal = {Nature},
  year   = {2015},
  volume = {521},
  number = {7553},              <- the issue number
  pages  = {436--444},
  doi    = {10.1038/nature14539} <- last field: comma optional
}
```

(The arrows are explanations for this guide only; do not type them.)

Rule	Example
The citation key must be unique and contain no spaces. A good pattern is author + year + first word of the title.	lecun2015deep, nist2020security
Every field except the last one ends with a comma.	year = {2015},
Separate authors with the word and, never with commas.	{LeCun, Yann and Bengio, Yoshua}
Write each name as Last, First.	Barto, Andrew G.
An organization as author needs double braces so it is not read as a person's name.	author = {{OpenAI}}
Protect capital letters that must stay capitalized (acronyms, proper nouns) with braces.	title = {Fine-tuning {BERT} for {Python} code}
Use two hyphens for page ranges.	pages = {436--444}
Write the DOI without the https://doi.org/ prefix; the style adds it.	doi = {10.1038/nature14539}

8.3 Entry Types for a Computing Project

Choose the entry type that matches the kind of source. These are the types you are most likely to need:

Source	Entry type	Key fields
Journal article	@article	author, title, journal, year, volume, number, pages, doi
Book	@book	author, title, edition, publisher, year
Chapter in an edited book	@incollection	author, title, editor, booktitle, publisher, year, pages
Conference paper	@inproceedings	author, title, booktitle, publisher, year, pages, doi
Web page or online documentation	@online	author, title, year (and month, day), url

Source	Entry type	Key fields
Software, library, or code repository	@software	author, title, version, year, url
Dataset	@dataset	author, title, titleaddon, publisher, year, url or doi
Technical report or standard	@report	author, title, number, institution, year, doi or url
Generative AI tool	@software	author, title, version, titleaddon, year, url

Here are complete examples you can copy and adapt:

```
@online{python_tutorial,
  author = {{Python Software Foundation}},
  title = {The {Python} tutorial},
  year = {2024},
  url = {https://docs.python.org/3/tutorial/}
}

@software{numpy_2024,
  author = {{NumPy Developers}},
  title = {{NumPy}},
  version = {2.1.0},
  year = {2024},
  url = {https://numpy.org}
}

@dataset{lecun1998mnist,
  author = {LeCun, Yann and Cortes, Corinna and
    Burges, Christopher J. C.},
  title = {The {MNIST} database of handwritten digits},
  titleaddon = {Data set},
  publisher = {Courant Institute, New York University},
  year = {1998},
  url = {http://yann.lecun.com/exdb/mnist/}
}

@report{nist2020security,
  author = {{National Institute of Standards and Technology}},
  title = {Security and privacy controls for information systems
    and organizations},
  number = {SP 800-53 Rev. 5},
  institution = {U.S. Department of Commerce},
  year = {2020},
  doi = {10.6028/NIST.SP.800-53r5}
}
```

These entries produce the following APA references:

Reference list output

Python Software Foundation. (2024). *The Python tutorial*. <https://docs.python.org/3/tutorial/>

NumPy Developers. (2024). *NumPy* (Version 2.1.0). <https://numpy.org>

Reference list output

LeCun, Y., Cortes, C., & Burges, C. J. C. (1998). *The MNIST database of handwritten digits* [Data set]. Courant Institute, New York University. <http://yann.lecun.com/exdb/mnist/>

National Institute of Standards and Technology. (2020). *Security and privacy controls for information systems and organizations* (SP 800-53 Rev. 5). U.S. Department of Commerce. <https://doi.org/10.6028/NIST.SP.800-53r5>

Tip: Retrieval dates

Add `urldate = {2026-10-01}` to an `@online` entry only when the page is designed to change over time (for example, a live dashboard or a frequently updated wiki page). APA then prints “Retrieved October 1, 2026, from ...”. For stable pages, leave it out.

8.4 Getting BibTeX Entries Quickly

You rarely need to type entries from scratch. Use one of these sources, then **check and clean** the result:

Source	How
Google Scholar	Search for the paper, click the quotation-mark icon (Cite) under the result, then click BibTeX . Copy the text into <code>references.bib</code> .
Publisher or library page	Look for Cite , Export citation , or Download citation and choose BibTeX .
A DOI	Paste the DOI into doi2bib.org to get a BibTeX entry.
Zotero (free reference manager)	Collect sources with the browser connector, then export the collection as BibTeX (the Better BibTeX add-on keeps keys stable).

Watch Out: Always check imported entries

Automatically generated entries are often incomplete or wrong. Before you cite one, check that:

- the **DOI** is present if the source has one (Google Scholar often leaves it out);
- acronyms and proper nouns in the **title** are protected with braces, for example `{BERT}`;
- the **entry type** is right (Scholar sometimes labels conference papers as articles); and
- you cite the **published version** of a paper rather than a preprint when one exists.

8.5 Citation Commands

You type	You get	Use it when
<code>\parencite{lecun2015deep}</code>	(LeCun et al., 2015)	The citation is in parentheses (most common).
<code>\textcite{lecun2015deep}</code>	LeCun et al. (2015)	The authors are part of your sentence.
<code>\parencite[p.~437]{lecun2015deep}</code>	(LeCun et al., 2015, p. 437)	You quote directly and need a page number.
<code>\parencite[para.~3]{python_tutorial}</code>	(Python Software Foundation, 2024, para. 3)	You quote a source without page numbers.
<code>\parencite{vaswani2017attention,lecun2015deep}</code>	(LeCun et al., 2015; Vaswani et al., 2017)	You cite several sources at once. They are sorted alphabetically for you.
<code>\parencite[see][]{lecun2015deep}</code>	(see LeCun et al., 2015)	You need a word before the citation. Note the empty second brackets.
<code>\citeauthor{lecun2015deep}</code>	LeCun et al.	You need only the author names.
<code>\citeyear{lecun2015deep}</code>	2015	You need only the year.

8.6 APA Citation Rules You Should Know

- **One or two authors:** both names appear every time: (Sutton & Barto, 2018), or Sutton and Barto (2018) in a sentence. APA uses “&” in parentheses and “and” in the sentence. The template handles this.
- **Three or more authors:** only the first author plus “et al.” from the first citation: (Vaswani et al., 2017). The template handles this.
- **Organization as author:** use the organization’s name: (National Institute of Standards and Technology, 2020).
- **No date:** leave out the `year` field and the citation shows “n.d.”: (Wichita State University, n.d.).
- **Same author, same year:** the template adds letters automatically: (Smith, 2020a, 2020b).
- **Paraphrase instead of quoting** whenever you can. A page number is optional for paraphrases and required for direct quotes.
- **Short quotations** (fewer than 40 words) go in quotation marks within the sentence, followed by the citation with a page number.
- **Long quotations** (40 words or more) go in a block quote without quotation marks, as shown below.

```
\begin{quote}
Text of the long quotation (40 words or more) goes here, without
quotation marks. The citation comes after the final punctuation.
\parencite[p.~437]{lecun2015deep}
\end{quote}
```

8.7 Citing Generative AI Tools

If you use a generative AI tool such as ChatGPT in a way your course allows, APA recommends crediting the company that makes the tool as the author, and describing in your text how you used it (for example, the prompt you entered). **Check the course policy on AI use with Prof. Cody Farlow before using these tools for any part of your report.**

```
@software{openai2023chatgpt,
  author    = {{OpenAI}},
  title     = {{ChatGPT}},
  version   = {Mar 14},
  titleaddon = {Large language model},
  year      = {2023},
  url       = {https://chat.openai.com/chat}
}
```

In the text, `\parencite{openai2023chatgpt}` produces (OpenAI, 2023). In the reference list, the entry appears as: OpenAI. (2023). *ChatGPT* (Version Mar 14) [Large language model]. <https://chat.openai.com/chat>. Change the version, year, and URL to match the tool and version you actually used. (APA’s own example writes the version as “Mar 14 version”; the small difference in wording comes from the citation software and is acceptable.)

8.8 The Reference List

The line `\printbibliography` creates the reference list. You never type it by hand. The template automatically:

- starts the list on a new page with the label “References” centered in bold;
- sorts entries alphabetically by the first author’s last name;
- uses double spacing and a 0.5-in. hanging indent; and
- turns DOIs into links of the form <https://doi.org/....>

Try It: 5 — Cite real sources (20 minutes)

1. Find one journal article or conference paper related to your project on Google Scholar. Copy its BibTeX entry into `references.bib`, then check and clean it (Section 8.4).
2. Add an `@software` entry for one library or framework your project uses (for example, React, Flask, or PyTorch).
3. In `main.tex`, cite the paper once with `\parencite` and once with `\textcite`.
4. Cite the software entry with `\parencite`.
5. Recompile and look at the reference list.

Checkpoint: Both sources appear in the reference list in APA format, sorted alphabetically, and your in-text citations show the author names and years (not bold citation keys or question marks).

CHAPTER 9

9. Tables

9.1 APA Table Rules

- The table **number** (Table 1, Table 2, ...) appears above the table in **bold**.
- The table **title** appears one line below the number in *italic title case*.
- Use **horizontal lines only**: one above the column headings, one below them, and one at the bottom. No vertical lines.
- A **note** below the table explains abbreviations, symbols, or the source of the data, and starts with *Note.* in italics.
- **Every table must be mentioned in the text** (for example, “as shown in Table 1”). Tables are numbered in the order you first mention them.

1	Table 1	1	Table number: bold (automatic)																
2	<i>Classification Performance of Each Model on the Test Set</i>	2	\caption{...}: italic, title case																
3	<table> <tr> <th>Model</th><th>Accuracy</th><th>Precision</th><th>Recall</th></tr> <tr> <td>Logistic regression</td><td>.82</td><td>.80</td><td>.79</td></tr> <tr> <td>Random forest</td><td>.87</td><td>.86</td><td>.85</td></tr> <tr> <td>CNN</td><td>.91</td><td>.90</td><td>.92</td></tr> </table>	Model	Accuracy	Precision	Recall	Logistic regression	.82	.80	.79	Random forest	.87	.86	.85	CNN	.91	.90	.92	3	Horizontal lines only (\toprule, \midrule, \bottomrule)
Model	Accuracy	Precision	Recall																
Logistic regression	.82	.80	.79																
Random forest	.87	.86	.85																
CNN	.91	.90	.92																
4	<i>Note.</i> Values are proportions. APA omits the leading zero for values that cannot exceed 1.	4	Note below the table (\figurenote{...})																

Figure 11. The sample table in the template and the code behind each part.

9.2 The Template Table, Line by Line

```

\begin{table}[htbp]
  \caption{Classification Performance of Each Model on the Test Set}
  \label{tab:results}
  {\centering
  \begin{tabular}{lccc}
    \toprule
    Model & Accuracy & Precision & Recall \\
    \midrule
    Logistic regression & .82 & .80 & .79 \\
    Random forest      & .87 & .86 & .85 \\
    CNN                 & .91 & .90 & .92 \\
    \bottomrule
  \end{tabular}}\par
  \vspace{0.5\baselineskip}
  \figurenote{Values are proportions. APA omits the leading zero
  for values that cannot exceed 1.}
\end{table}

```


Code	Meaning
<code>\begin{table}[htbp]</code>	Starts a table <i>float</i> . The letters list the positions LaTeX may use: h ere, t op of a page, b ottom, or a separate p age.
<code>\caption{...}</code>	The table title. The template adds “Table 1” in bold above it and sets the title in italics.
<code>\label{tab:results}</code>	A name you can refer to with <code>Table~\ref{tab:results}</code> . Always put it <i>after</i> <code>\caption</code> .
<code>{\centering ... \par}</code>	Centers the table body only, so the title and note stay flush left as APA requires.
<code>\begin{tabular}{lccc}</code>	The table body. <code>lccc</code> means four columns: one left-aligned and three centered.
<code>&</code> and <code>\\</code>	<code>&</code> separates columns; <code>\\</code> ends a row.
<code>\toprule</code> , <code>\midrule</code> , <code>\bottomrule</code>	The three APA horizontal lines.
<code>\figurenote{...}</code>	The note below the table, starting with <i>Note</i> .

9.3 Column Types

Specifier	Column type
<code>l</code>	Left-aligned text. Good for labels and names.
<code>c</code>	Centered. Good for short values.
<code>r</code>	Right-aligned. Good for numbers you want lined up.
<code>p{6cm}</code>	A paragraph column of fixed width that wraps long text. Good for requirements and descriptions.

9.4 Example: A Requirements Table

Long text needs a `p{...}` column so it wraps. `\small` makes the font slightly smaller to fit more text.

```

\begin{table}[htbp]
  \caption{Functional Requirements of the SmartPark System}
  \label{tab:requirements}
  {\centering\small
  \begin{tabular}{l p{8cm} c}
    \toprule
    ID & Requirement & Priority \\
    \midrule
    FR-1 & The system shall show the number of open spaces in each
           parking lot, updated at least every 60 seconds. & High \\
    FR-2 & Users shall be able to filter parking lots by permit type. & Medium \\
    FR-3 & The system shall send an alert when a saved lot is full. & Low \\
    \bottomrule
  \end{tabular}\par}
\end{table}

```

Watch Out: Count your ampersands

Every row must have the same number of & signs: one fewer than the number of columns. Too many causes the error “Extra alignment tab has been changed to \cr.”

Tip: Building tables faster

- [TablesGenerator.com](https://www.tablesgenerator.com) converts a table you paste from Excel or Google Sheets into LaTeX. Turn on the **booktabs** option so it uses the APA-style lines, then copy only the tabular part into the template’s table structure.
- Keep tables simple. If a table is too wide for the page, shorten the column headings, use \small, or split it into two tables.

CHAPTER 10

10. Figures

Figures include charts, diagrams, screenshots, photographs, and any other images. APA formats figures the same way as tables: bold number, italic title, image, and an optional note.

10.1 Uploading an Image

1. In the Overleaf file tree, click the `figures` folder to select it.
2. Click the **Upload** icon at the top of the file tree and drag your image into the window, or select it from your computer.
3. The image appears inside `figures`. Its path is now `figures/your-file-name.png`.

Format	Best for
PDF (vector)	Charts and diagrams exported from Python, R, draw.io, or PowerPoint. They stay sharp at any size.
PNG	Screenshots of your application and diagrams with sharp edges.
JPG	Photographs.

Watch Out: File names

File names on Overleaf are case-sensitive and should not contain spaces. Name files like `system-architecture.pdf` or `login_screen.png`. `Login Screen.PNG` will cause problems.

10.2 The Figure Code

```
\begin{figure}[htbp]
  \caption{Training and Validation Accuracy Across 20 Epochs}
  \label{fig:accuracy}
  {\centering
    \includegraphics[width=0.75\linewidth]{figures/sample-figure.png}\par}
  \figurenote{Replace this figure with your own. Use the note to explain
    abbreviations or to credit the source of an adapted figure.}
\end{figure}
```

Code	Meaning
<code>\caption{...}</code>	The figure title, in title case. The template adds “Figure 1” in bold above it.
<code>\label{fig:accuracy}</code>	Lets you write <code>Figure~\ref{fig:accuracy}</code> in the text. Always place it after <code>\caption</code> .

Code	Meaning
<code>width=0.75\linewidth</code>	Makes the image 75% of the text width. Use <code>1\linewidth</code> for full width and <code>0.5\linewidth</code> for half.
<code>figures/sample-figure.png</code>	The path to your image. It must match the file name exactly, including capital letters.
<code>\figurenote{...}</code>	Optional note below the figure. Delete the line if you do not need it.

10.3 Referring to Tables, Figures, and Other Elements

Never type a number like “Figure 3” by hand. Use `\label` and `\ref` so that numbers update automatically when you add or move things.

Element	Label it	Refer to it
Table	<code>\label{tab:requirements}</code>	<code>Table~\ref{tab:requirements}</code>
Figure	<code>\label{fig:architecture}</code>	<code>Figure~\ref{fig:architecture}</code>
Equation	<code>\label{eq:accuracy}</code>	<code>Equation~\ref{eq:accuracy}</code>
Appendix	<code>\label{app:manual}</code>	<code>Appendix~\ref{app:manual}</code>

The prefixes `tab:`, `fig:`, `eq:`, and `app:` are not required, but they make your labels easy to read and keep them from clashing. If you see “??” in the PDF, recompile once more; if the question marks stay, check that the label name is spelled the same way in both places.

10.4 Why Did My Figure Move?

Tables and figures are *floats*: LaTeX places them where they fit best, which may be the top of the next page. This is normal and acceptable in APA papers, because you refer to them by number. Do not fight it with blank lines or `\vspace`. If a float drifts too far, try `[!htbp]` instead of `[htbp]`, or move the figure code a paragraph earlier in your source.

10.5 Figures You Did Not Create

If you reproduce or adapt a figure from another source, you must credit it in the figure note and include the source in your reference list. A typical note looks like this:

```
\figurenote{Adapted from \enquote{Title of the Article,} by A. A. Author
and B. B. Author, 2020, \textit{Journal Name}, \textit{12}(3), p. 45
(https://doi.org/xx.xxxx). Copyright 2020 by the Copyright Holder.}
```

Images you create yourself, such as screenshots of your own application or diagrams you drew, need no credit.

10.6 Diagrams for Software Projects

- **Architecture, UML, and flow diagrams:** draw them in [diagrams.net](https://draw.io) (draw.io) and export as PDF or PNG.
- **Charts of results:** create them in Python (matplotlib), R, or Excel and export as PDF or PNG at 300 dpi.
- **Screenshots:** capture your application at its normal size. Crop out anything unrelated, such as browser tabs or personal information.

Try It: 6 — Add a table and a figure (20 minutes)

- Add a table that lists three functional requirements of your project, using the example in Section 9.4.
- Upload one image related to your project (a screenshot, sketch, or diagram) to `figures` and insert it as a figure.
- Give both a `\label` and refer to both in a sentence using `\ref`.

Checkpoint: Your PDF shows “Table 1” and “Figure 1” (or the next numbers) with italic titles, and your sentence shows the correct numbers instead of “??”.

CHAPTER 11

11. Equations, Code, Appendices, and More

11.1 Equations

Use inline math for short expressions within a sentence, and the `equation` environment for important formulas you want to number and refer to. The template includes an example (the F1 score).

```
We evaluate each model with the F1 score, defined in Equation~\ref{eq:f1}:
\begin{equation}
  F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}
  \label{eq:f1}
\end{equation}
where precision and recall are computed on the test set.
```

Treat equations as part of the sentence: continue the sentence after the equation and do not start a new paragraph unless you mean to.

11.2 Code in Your Report

APA has no special rules for source code. Good practice for a Senior Project report:

- Include only **short, important excerpts**, such as a key algorithm, in the text or in an appendix.
- **Link to your full repository** instead of pasting hundreds of lines, and cite it as an `@software` entry.
- Explain what the excerpt does in your own words.

To format code nicely, add these two lines to the preamble (in the packages block):

```
\usepackage{listings}
\lstset{basicstyle=\ttfamily\small, breaklines=true,
        frame=single, columns=fullflexible}
```

Then include code like this:

```
\begin{lstlisting}[language=Python]
def f1_score(precision, recall):
    return 2 * precision * recall / (precision + recall)
\end{lstlisting}
```

Common language names include `Python`, `Java`, `C`, `C++`, `SQL`, and `HTML`. For a language that `listings` does not know, simply leave out `[language=...]`.

11.3 Appendices

Appendices come after the reference list. Each one starts on a new page. Start them with `\appendix`; after that, every `\section` becomes a new appendix labeled A, B, C, and so on.

```
\printbibliography

\appendix
\section{User Manual}
\label{app:manual}
...

\section{Test Cases}
\label{app:tests}
...
```

In the text, write `see Appendix~\ref{app:manual}`. Typical Senior Project appendices include a user manual, installation instructions, detailed test cases, the full list of requirements, meeting notes, and a team contribution statement.

11.4 Links, Footnotes, and Other Small Elements

You need	Use	Notes
A web address	<code>\url{https://github.com/team/repo}</code>	Handles special characters and line breaks in long URLs.
Linked text	<code>\href{https://www.wichita.edu}{university website}</code>	Shows the text; clicking it opens the link.
A footnote	<code>text\footnote{Extra detail.}</code>	APA discourages footnotes; use them rarely.
A non-breaking space	<code>Table~1, Section~3</code>	Keeps the word and number on the same line.
A new page	<code>\clearpage</code>	Rarely needed; avoid it unless necessary.

PART III · YOUR REPORT

12. Writing Your Senior Project Report

Note: Your assignment instructions come first

This chapter suggests a structure that is common in computing capstone reports. **The official assignment instructions and rubric from Prof. Cody Farlow always take priority over anything in this guide.** If they ask for different sections, use theirs.

12.1 Senior Project I and Senior Project II

The two semesters usually emphasize different parts of the project. A typical split looks like this:

	Senior Project I (typical focus)	Senior Project II (typical focus)
Main question	What are we building, and why?	What did we build, and how well does it work?
Typical content	Problem statement, background and related work, requirements, proposed design, project plan and timeline, risks	Implementation, testing and evaluation, results, discussion of limitations, conclusions, user documentation
Typical figures and tables	Requirements table, architecture diagram, UI mockups, Gantt chart	Screenshots, test results tables, performance charts

12.2 A Suggested Report Outline

Section (heading)	What goes in it	LaTeX
Title page	Title, team members, affiliation, course, instructor, due date	Title-page block
Abstract (if required)	One-paragraph summary of the problem, approach, and results	<code>\abstract{...}</code>
(Introduction, no heading)	The problem, why it matters, who it affects, project objectives, and scope	Text after <code>\maketitle</code>
Background and Related Work	Existing solutions, relevant technologies, and research; what is missing that your project provides. Cite sources here.	<code>\section</code>
Requirements	Functional and non-functional requirements, ideally in a numbered table	<code>\section</code> + table
System Design	Architecture, main components, data model, interface design, key design decisions and trade-offs	<code>\section</code> + figures

Section (heading)	What goes in it	LaTeX
Implementation	Technologies and why you chose them, important algorithms, challenges and how you solved them	<code>\section</code>
Testing and Evaluation	Test plan, test types (unit, integration, user testing), results, metrics	<code>\section</code> + tables
Discussion	What worked, what did not, limitations, ethical, privacy, and security considerations	<code>\section</code>
Project Management	Timeline, roles, team contributions, how you collaborated	<code>\section</code>
Conclusion and Future Work	Main outcomes and what could be done next	<code>\section</code>
References	Generated automatically	<code>\printbibliography</code>
Appendices	User manual, test cases, extra material	<code>\appendix</code> + <code>\section</code>

12.3 The Outline in LaTeX

You can paste this skeleton into the body of `main.tex` to start your report:

```

\maketitle

% Introduction (no heading): the problem, motivation, objectives, scope.

\section{Background and Related Work}

\section{Requirements}
\subsection{Functional Requirements}
\subsection{Non-Functional Requirements}

\section{System Design}
\subsection{Architecture}
\subsection{Data Model}
\subsection{User Interface}

\section{Implementation}

\section{Testing and Evaluation}
\subsection{Test Plan}
\subsection{Results}

\section{Discussion}

\section{Project Management}

\section{Conclusion and Future Work}

\printbibliography

\appendix
\section{User Manual}

```

12.4 What to Cite in a Software Project

In a software project, many sources besides research papers need credit. Cite:

- research papers, books, and articles whose ideas, methods, or results you use;
- major **libraries and frameworks** your system depends on (as `@software`);
- **datasets** you use for training, testing, or analysis (as `@dataset`);
- **APIs and external services** your system calls (their documentation, as `@online`);
- **code you adapted** from tutorials, Stack Overflow, or other repositories (and respect their licenses);
- **images, icons, and figures** you did not create;
- **standards and guidelines** you followed, such as accessibility or security standards (as `@report` or `@online`); and
- **generative AI tools**, if the course policy allows their use (Section 8.7).

Key Idea: Academic integrity

Presenting someone else's words, code, images, or ideas as your own is plagiarism, even if it is accidental. When in doubt, cite. A source that is cited correctly never hurts your report.

12.5 APA Writing-Style Essentials

Topic	APA guidance	Example
Person and voice	First person is fine for describing what your team did. Prefer active voice.	"We designed a REST API" rather than "A REST API was designed."
Verb tense	Past tense for what you did and what you found; present tense for what the results mean and for describing how the system works.	"We tested 50 users. The results suggest that..."
Numbers	Words for zero to nine; numerals for 10 and above. Always use numerals with units, percentages, times, and statistics.	"three modules," "12 modules," "5 ms," "4%"
Abbreviations	Spell out the full term on first use with the abbreviation in parentheses; then use the abbreviation.	"application programming interface (API)"
Lists in a sentence	Use lowercase letters in parentheses.	"The system (a) collects, (b) cleans, and (c) stores data."
Bias-free language	Be specific and respectful. Use "they" for a person whose gender is unknown or irrelevant.	"Each user can update their profile."
Clarity	Short, direct sentences. Define technical terms the reader may not know. Avoid filler and slang.	"The cache reduces load time by 40%."

12.6 Writing as a Team

Because the free Overleaf plan allows one invited collaborator per project (Section 2.3), plan how your team will write before you start:

Option	How it works	Good for
Share in Overleaf	One teammate owns the project and invites one collaborator by email (Share button). Others can be given a view-only link.	Teams of two
Rotate the editor	One person types into Overleaf at a time; others draft in shared notes and hand over text.	Short writing sessions
One file per section	Each teammate writes a section file (Section 6.7) and sends it to the project owner, who uploads it.	Larger teams on the free plan

Option	How it works	Good for
Git and GitHub	Keep the report in your project's Git repository and compile on your own computers (Appendix B). No collaborator limit.	Teams already comfortable with Git

- **Agree on conventions early:** citation key format (for example, `authorYYYYword`), label prefixes, and file names.
- **Assign one reference manager** who checks `references.bib` for duplicates and errors before each submission.
- **Leave notes as comments**, such as `% TODO (Sam): add test results`.
- **Compile often**, so errors are found when they are small.
- **Back up regularly:** download the source with **File → Download → Download as source (.zip)**, because the free plan keeps only recent history.

12.7 Submission Checklist

Before you submit, check each item:

✓	Check
☐	The project compiles with no errors (the logs show no red error count).
☐	The title page has the real title in title case, all team members, the affiliation, the course, Prof. Cody Farlow, and the due date.
☐	The abstract is included only if the assignment requires it, and it fits the word limit.
☐	There is no “Introduction” heading, and headings follow the APA levels without skipping.
☐	All sample text, sample tables, sample figures, and the sample appendix from the template have been removed or replaced.
☐	Every table and figure has a title, is mentioned in the text, and is numbered correctly.
☐	No “??” appears anywhere in the PDF, and no citation key appears in bold.
☐	Every source you used is cited, and every reference list entry is cited in the text.
☐	Reference entries are complete (authors, year, title, source, DOI or URL) and acronyms keep their capitals.
☐	Direct quotations have quotation marks (or block format) and page or paragraph numbers.
☐	Adapted figures have a credit note.
☐	Numbers, abbreviations, and tense follow Section 12.5, and the spell check is clean.
☐	Appendices (if any) are referred to in the text.
☐	You downloaded the final PDF, opened it, and checked every page.

✓	Check
☐	The file is named and submitted as the assignment instructions require.

Try It: 7 — Build your report skeleton (30 minutes)

1. Replace the template's sample body with the outline from Section 12.3, adjusted to your assignment instructions.
2. Write two or three sentences of introduction under `\maketitle`.
3. Add at least three real sources to `references.bib` and cite each one.
4. Add one table and one figure from your project.
5. Go through the submission checklist and fix anything that does not pass.

Checkpoint: You have a compiling report skeleton with your title page, headings, citations, a table, a figure, and a reference list. From here, you only need to write.

CHAPTER 13

13. Troubleshooting and FAQ

13.1 A Debugging Routine

1. **Open the logs** (the icon next to **Recompile**) and read the **first** error.
2. **Go to the line** it mentions. Click the error to jump there. Check that line and the one before it.
3. **Undo your last change** if you are not sure what caused the error, and recompile.
4. **Narrow it down.** Comment out half of the recent text with `%`. If the error disappears, the problem is in that half. Repeat.
5. **Clear the cache.** If the PDF looks out of date or references are stuck, open the logs and click **Clear cached files** at the bottom, then recompile.

Key Idea: Errors and warnings are different

Errors (red) must be fixed. **Warnings** (yellow), such as “Overfull \hbox,” usually mean something small, like a line that sticks out slightly. You can often ignore warnings, but look at the PDF to make sure.

13.2 Common Errors and How to Fix Them

Error message (or symptom)	What it usually means	How to fix it
Undefined control sequence	A command is misspelled, has the wrong capitalization, or needs a package.	Check the spelling and case of the command on that line. Add the package if needed.
Missing \$ inserted	A math-only character such as <code>_</code> or <code>^</code> is used in normal text.	Write <code>_</code> in text, or put the math inside <code>\$. . . \$</code> .
Missing } inserted / Runaway argument / File ended while scanning	A brace <code>{</code> was never closed.	Find the opening brace and add the matching <code>}</code> .
<code>\begin{...}</code> ended by <code>\end{...}</code>	An environment was not closed, or the names do not match.	Make sure every <code>\begin{x}</code> has a matching <code>\end{x}</code> .
File <code>figures/x.png</code> not found	Wrong file name or path, wrong capitalization, or the file was not uploaded.	Compare the name in your code with the file tree, letter by letter.
Extra alignment tab has been changed to <code>\cr</code>	A table row has too many <code>&</code> .	Each row needs one fewer <code>&</code> than the number of columns.
Misplaced alignment tab character <code>&</code>	<code>&</code> is used in normal text.	Write <code>\&</code> .

Error message (or symptom)	What it usually means	How to fix it
Bold citation key in the PDF, or “Citation ... undefined”	The key is misspelled or the entry is missing or broken in <code>references.bib</code> .	Check the key in both files. Check the entry for a missing comma or brace. Recompile.
References page is empty	Nothing is cited yet, or Biber failed because of an error in <code>references.bib</code> .	Cite at least one source. Look for Biber messages in the logs.
“??” instead of a number	A <code>\ref</code> points to a label that does not exist yet, or needs another compile.	Recompile. Check that the label is spelled the same in <code>\label</code> and <code>\ref</code> .
Unicode character ... not set up for use with LaTeX	You pasted an unusual character (an emoji, a special symbol, or an invisible character from Word or a PDF).	Delete and retype the text around the line mentioned.
Text after % disappears	% starts a comment.	Write <code>\%</code> .
Compile timeout	The project took longer than the free plan allows, often because of very large images or a chain of errors.	Fix errors first. Compress large images. Use PDF or PNG figures at a reasonable size.
Overfull \hbox (warning)	A line is slightly too wide, often a long URL or word.	Usually ignore it. Wrap URLs in <code>\url{...}</code> .

13.3 Problems With References

- **A missing comma** between fields in `references.bib` is the most common cause of broken references. Check the entry you added most recently.
- **Duplicate keys:** two entries with the same key confuse Biber. Each key must be unique.
- **Capital letters disappear** in a title (for example, “Bert” instead of “BERT”): protect them with braces, `{BERT}`.
- **A person’s name looks wrong:** check that authors are separated by `and` and written as `Last, First`.
- **An organization’s name is shortened to initials:** use double braces, `{{National Institute of Standards and Technology}}`.

13.4 Frequently Asked Questions

Q: Can I write my report in Word or Google Docs instead?

Follow the requirements in your assignment instructions. This guide covers the LaTeX template.

Q: Do I need an abstract?

Only if your assignment asks for one. APA does not require it for student papers.

Q: Do I need a running head?

No. APA 7 student papers do not have a running head, and the template leaves it out.

Q: Can I change the font?

Keep the template font unless your instructor asks otherwise. It matches the APA Times New Roman 12-pt option.

Q: How do I count words?

In Overleaf, choose **File** → **Word Count**. It counts text and ignores LaTeX commands.

Q: How do I download my PDF?

Click the **Download PDF** icon next to **Recompile**, or choose **File** → **Download** → **Download as PDF**.

Q: How do I download my source files?

Choose **File** → **Download** → **Download as source (.zip)**. Do this regularly as a backup.

Q: I deleted something important. Can I get it back?

Try **History** (top right). The free plan keeps only recent history, which is why regular source backups are a good idea.

Q: How do I share my project with Prof. Farlow?

Submit the PDF as the assignment instructs. If you are asked to share the project, use **Share** to create a view-only link.

Q: Why are my figures on a different page than the text?

Figures and tables are floats (Section 10.4). This is normal.

Q: The template was updated. How do I get the new version?

Download the new zip from the repository, upload it as a new project, and copy your text, `references.bib` entries, and figures into it.

Q: Can I use this template for other courses?

Yes, for non-commercial academic work, under the CC BY-NC 4.0 license. Check each instructor's requirements first.

APPENDICES

Appendix A. Quick Reference Cheat Sheet

Document and Structure

Task	Code
Title page fields	<code>\title{} \authorsnames{} \authorsaffiliations{[]} \course{} \professor{} \duedate{}</code>
Headings, Levels 1–5	<code>\section{} \subsection{} \subsubsection{} \paragraph{} \subparagraph{}</code>
New paragraph	Leave a blank line
Comment	<code>% this is hidden</code>
Include another file	<code>\input{sections/design}</code>
Appendices	<code>\appendix then \section{Title}</code>

Text

Task	Code	Task	Code
Bold	<code>\textbf{text}</code>	Quotation marks	<code>``text''</code> or <code>\enquote{text}</code>
Italic / emphasis	<code>\textit{text}</code> <code>\emph{text}</code>	En dash (ranges)	<code>10--20</code>
Monospace	<code>\texttt{text}</code>	Em dash	<code>---</code>
Bullet list	<code>\begin{itemize} \item ...</code>	Non-breaking space	<code>Table~1</code>
Numbered list	<code>\begin{enumerate} \item ...</code>	URL / link	<code>\url{...}</code> <code>\href{url}{text}</code>

Special Characters

%	&	_	#	\$	{ }	~	^	\
<code>\%</code>	<code>\&</code>	<code>_</code>	<code>\#</code>	<code>\\$</code>	<code>\{ }</code>	<code>\textasciitilde{}</code>	<code>\textasciicircum{}</code>	<code>\textbackslash{}</code>

Citations

Code	Output
<code>\parencite{key}</code>	(Author, 2020)
<code>\textcite{key}</code>	Author (2020)
<code>\parencite[p.~12]{key}</code>	(Author, 2020, p. 12)
<code>\parencite{key1,key2}</code>	(Author A, 2019; Author B, 2020)
<code>\parencite[see][key]</code>	(see Author, 2020)

Code	Output
<code>\citeauthor{key} / \citeyear{key}</code>	Author / 2020

Tables, Figures, and Equations

```

\begin{table}[htbp]
  \caption{Title in Title Case}
  \label{tab:name}
  {\centering
  \begin{tabular}{lcc}
    \toprule
    A & B & C \\
    \midrule
    1 & 2 & 3 \\
    \bottomrule
  \end{tabular}}\par
  \figurenote{Optional note.}
\end{table}

\begin{figure}[htbp]
  \caption{Title in Title Case}
  \label{fig:name}
  {\centering
  \includegraphics[width=0.75\linewidth]
    {figures/file.png}}\par
  \figurenote{Optional note.}
\end{figure}

\begin{equation}
  E = mc^2 \label{eq:name}
\end{equation}

See Table~\ref{tab:name}.

```

Overleaf Shortcuts

Action	Windows	Mac
Recompile	Ctrl + Enter	Cmd + Enter
Comment / uncomment	Ctrl + /	Cmd + /
Bold / italic	Ctrl + B / Ctrl + I	Cmd + B / Cmd + I
Find and replace	Ctrl + F	Cmd + F
Jump from PDF to source	Double-click in the PDF	Double-click in the PDF

APPENDIX B

Appendix B. Working on Your Own Computer (Optional)

Overleaf is all you need for this course. Working locally is useful if you want to write offline, use your own editor, or keep the report in a Git repository with your code, with no limit on collaborators.

B.1 Install a LaTeX Distribution

System	Distribution	Notes
Windows	MiKTeX or TeX Live	MiKTeX installs missing packages on demand; turn on Always install missing packages during setup. To use <code>latexmk</code> with MiKTeX you also need Perl (for example, Strawberry Perl). TeX Live includes everything but is a large download.
macOS	MacTeX	A complete TeX Live distribution for Mac. Large download; allow time.
Linux	TeX Live	Install from your package manager (for example, the <code>texlive-full</code> package on Ubuntu) or from tug.org .

The template needs the `apa7`, `biblatex-apa`, `newtx`, and `csquotes` packages and the **Biber** program. Full TeX Live and MacTeX installations include all of them.

B.2 Choose an Editor

Editor	Setup
Visual Studio Code with the LaTeX Workshop extension	Install the extension from the Extensions panel. Its default recipe uses <code>latexmk</code> , which runs Biber automatically. Save the file to build, and open the PDF in a tab beside your code.
TeXstudio	A dedicated LaTeX editor. In Options → Configure TeXstudio → Build , set Default Bibliography Tool to Biber .

B.3 Compile From the Command Line

```
latexmk -pdf main.tex      # builds main.pdf; runs pdfLaTeX and Biber as needed
latexmk -c                 # removes temporary files, keeps the PDF
```

Without `latexmk`, run the steps from Figure 2 yourself: `pdflatex main`, then `biber main`, then `pdflatex main` twice.

B.4 Get the Template With Git

Install [Git](#) (or [GitHub Desktop](#) if you prefer a graphical app), then clone the repository:

```
git clone https://github.com/ParisaRostaami/APA7-LaTeX-Template.git
cd APA7-LaTeX-Template/template
```

The `template` folder contains the same files as the zip. Copy it into your own team repository (for example, as a `report` folder next to your code). The repository also includes a `.gitignore` file that keeps LaTeX's temporary files out of Git; copy it too.

B.5 A Simple Team Workflow With Git

1. **Pull before you write:** `git pull`.
2. **Write and compile** your section.
3. **Commit small, focused changes:** `git add report/sections/design.tex`, then `git commit -m "Add architecture section"`.
4. **Push:** `git push`.

Tip: One sentence per line

LaTeX ignores single line breaks, so you can start each sentence on a new line. Git then shows changes sentence by sentence, and two teammates editing the same paragraph rarely conflict.

APPENDIX C**Appendix C. Hands-On Exercises at a Glance**

These exercises follow the order of the guide and are designed to be done in class, step by step. Each one builds on the previous one.

#	Exercise	Section	Time	You will have
1	Your first LaTeX document	2.5	10 min	An Overleaf account and a compiled practice document
2	Practice the basics	3.10	15 min	Special characters, lists, equations, and one fixed error
3	Set up your report project	5.4	10 min	Your own copy of the template, compiled in Overleaf
4	Make the title page yours	6.7	10 min	A correct APA title page for your team
5	Cite real sources	8.8	20 min	Two correctly cited sources in an APA reference list
6	Add a table and a figure	10.6	20 min	An APA table and figure, referenced in the text
7	Build your report skeleton	12.7	30 min	A compiling skeleton of your actual report

Total: about 2 hours. Exercises 1–4 fit a first session, and Exercises 5–7 fit a second one.

APPENDIX D

Appendix D. Further Resources

The Template

- Template repository: github.com/ParisaRostaami/APA7-LaTeX-Template
- The `apa7` LaTeX class (documentation): ctan.org/pkg/apa7
- The `biblatex-apa` citation style (documentation): ctan.org/pkg/biblatex-apa

APA Style

- Purdue OWL, APA General Format: owl.purdue.edu
- APA Style official website: apastyle.apa.org
- APA Style blog, citing ChatGPT: apastyle.apa.org/blog/how-to-cite-chatgpt

LaTeX and Overleaf

- Overleaf, Learn LaTeX in 30 Minutes: overleaf.com/learn
- Overleaf documentation: docs.overleaf.com
- Detexify (find a symbol by drawing it): detexify.kirelabs.org
- Tables Generator: tablesgenerator.com

Tools for Your Project

- Zotero, a free reference manager: zotero.org
- doi2bib, BibTeX from a DOI: doi2bib.org
- diagrams.net (draw.io), free diagrams: diagrams.net

Note: About this guide and template

The APA 7 LaTeX template and this guide were created by **Parisa Rostami**, Graduate Teaching Assistant, for Senior Project I & II (Prof. Cody Farlow), School of Computing, Wichita State University. The template builds on the [apa7](#) class by Daniel A. Weiss and the [biblatex-apa](#) style by Philip Kime.

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Found a problem or have a suggestion? Open an issue in the GitHub repository or tell your GTA.